DAY-05.

Type conversion In JAVA :

Data Types
- Integers (Byte, short, int, long)
- Real (float, double)
- Character char
- Booleane. boolean

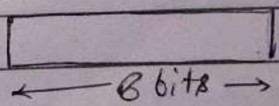
(1) Type conversion in Java
- → Implicit
- Explicit.

```
byte b = 24;
 int i ;
  i = b;
System.out.println(b) //24.
```

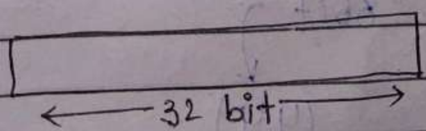
★ Rule for Implicit conversion.

Distination data type should be wider then source data type.

(1) byte → -128 to 127
  8 bit



(2) int - 32 bit



Widening Conversion.

short → int
int → long

② Explicit conversion.
  └ casting

```
int i = 1024
   or
int i = 300;
byte b;
    b = i;   — error at this line
b = (byte)(i); ✓   we casting.

System.out.println(b);   // 44
```

Q. How?

int i = 300;

2	300	
2	150	0
2	75	0
2	37	1
2	18	1
2	9	0
2	4	1
2	2	0
2	1	0
	0	1

LSB

000.. 000 0000 | 00101100

8bit 8bit 8bit 8bit

$\boxed{00101100}$ byte

8bit

(44)

★ Alternat ways to calculate result after cast.

b = 300 % (Range of byte) $2^8 \rightarrow 256$

300 % 256

$\boxed{b = 44}$

long → int
double → Float > Narrowing conversion.

→ char c = 'a';   char = 16bit
  int i = c;   int = 32bit

System.out.println(i);   // 97.

③ Truncating conversion.

Float → int
double → int

```
Float b = 16.25f;
  int i;                (truncate decimal part)
     i = b;  ×
  i = (int)b;      // i = 16.
```

→ boolean to any data type.
These conversions are not possible.

⇒ Automatic Type promotion.

```
byte b = 50;
byte a = 40;
byte c = 100;

int i = (a*b)/c;
```

$\underbrace{(a*b)}$ This calculation perform in term of int data type.

as

50*40 → 2000 — out of byte Range
int

2000/100 → 200

# byte b = 50;
b = b*2;
→ 50*2 → 100 (int)
(int)

fix as.
b = (byte)(b*2);

Rules of Type promotion.

① byte, short and char values are promoted to int.

② If one operand is long, the entire whole expression will become long.

③ If one operand is float, entire expression will become float.

④ If one operand is double, entire expression is double.